Vision-Assisted Smart Irrigation System for Crop Specific Water Management

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1. Abstract

Water scarcity and the use of inefficient irrigation methods are still major problems in modern agriculture. Traditional irrigation is mostly based on manual operation or timers and as a result often causes overwatering, under watering and the waste of resources. This project introduces an IoT-enabled smart irrigation and crop monitoring system which automates the decision to water by using real-time environmental data. To evaluate the conditions in the field, the system makes use of a soil moisture sensor, a rain sensor and a DHT11 temperature-humidity sensor, while a PIR motion sensor is used to watch for any intrusion or animal movement. The ESP32-S3 microcontroller analyses the sensor data and controls a water pump via a 2-channel relay module in such a way that watering only takes place when it is actually required. The system includes no physical display; all monitoring and control are carried out remotely via the Blynk IoT app, the app showing live gauges for soil moisture, rain, PIR, temperature and humidity together with an indicator of the pump's status. It also has buttons which allow the user to manually turn the water pump on or off according to the readings displayed and to immediately deal with any motion-detection alert from the field. In addition to the automation based on sensors, the system has an image-based crop health monitoring process: a crop image taken with a smartphone is sent to a server for inference using Edge Impulse and the result is stored in a Firebase database, which the ESP32-S3 checks in order to help make irrigation and alert decisions. The combination of these features results in a low-cost, scalable and water-efficient solution for smart farming.

Keywords – *ESP32-S3, Soil Moisture Sensor, DHT11, PIR Sensor, Rain Sensor, Blynk IoT, Edge Impulse, Firebase, Smart Irrigation*

2. Introduction

2.1 Background

Agriculture continues to be one of the largest consumers of fresh water in the world, and inefficient irrigation practices remain a major contributor to water wastage [1]. Conventional irrigation is largely dependent on manual observation or fixed timers, neither of which accounts for the actual condition of the soil, the crop, or the surrounding weather at a given moment. As a result, fields are frequently overwatered or under watered, leading to reduced crop yield, higher water bills and, over time, degradation of soil health [1].

The Internet of Things (IoT) has enabled sensors, actuators and microcontrollers to be networked together so that environmental information can be collected continuously, processed intelligently and acted upon in real time [1], [2]. When combined with cloud platforms and mobile dashboards, IoT allows a farmer to monitor and control the irrigation process from anywhere, without needing to be physically present in the field [5].

This project presents a Vision-Assisted Smart Irrigation System that goes a step further than conventional sensor-only automation. In addition to soil moisture, rainfall, temperature, humidity and intrusion sensing, the system incorporates an image-based crop health assessment pipeline. A photograph of the crop captured on a smartphone is analyzed using an Edge Impulse machine-learning model, and the resulting classification is stored on a Firebase database that the ESP32-S3 consults before deciding whether, and how much, to irrigate. This fuses conventional sensor-driven automation with lightweight computer vision to make the irrigation decision genuinely crop-specific.

2.2 Motivation

Water is a finite resource, and the pressure on it is expected to increase as agricultural demand grows [1]. A system that can water a field only when the soil, the weather and the visible condition of the crop indicate a genuine need for it can meaningfully reduce water consumption while also improving yield consistency.

The motivation behind this project is to design a low-cost, compact and largely display-free irrigation controller, built around commonly available sensors and the ESP32-S3, that can be monitored and operated entirely through a mobile application. By adding an inexpensive vision-based crop-health check on top of the sensor network, the system aims to move a step closer to precision, crop-specific irrigation without requiring expensive dedicated agricultural hardware.

3. Literature Survey

A number of studies have investigated IoT-based irrigation systems that rely on ESP32/ESP8266 class microcontrollers together with soil-moisture, temperature and rainfall sensors to automate watering decisions [3], [4], [7]. Obaideen et al. reviewed the contribution of smart, IoT-enabled irrigation to the United Nations Sustainable Development Goals, particularly Goal 6 on clean water, and highlighted how sensor-driven automation reduces water wastage compared with manual or timer-based irrigation [1].

Morchid et al. proposed a layered architecture in which an ESP32 device layer collects soil moisture, temperature and water-level data and communicates it in real time to a web interface using WebSockets, allowing remote visualization and control of the irrigation process [2]. Similar mobile-integrated and cloud-connected irrigation monitoring systems have been reported by Vaishali et al. and Saraf and Gawali, both of which use IoT dashboards for remote farm monitoring [11], [12]. Rajkumar et al. presented an early IoT-based intelligent irrigation approach that switched a pump automatically based on soil-moisture thresholds [10].

More recent work has combined irrigation automation with machine learning. Tace et al. combined IoT sensing with machine-learning-based prediction to optimize irrigation scheduling and reported measurable water savings [5]. Gamal et al. surveyed a wide range of smart-irrigation architectures and control techniques, noting a general trend toward combining wireless sensing with decision-support algorithms [6]. Kunt developed an ESP32 and Blynk based irrigation system that used an LSTM model for autonomous watering decisions, employing rain and soil-moisture sensors, a DHT11 sensor and relay-driven valves very similar to the sensor set used in this project [3]. Kwok and Sun proposed an IoT irrigation system that used deep-learning-based plant recognition to decide on watering, an early example of vision-assisted irrigation [4].

On the computer-vision side, Edge Impulse has demonstrated that lightweight convolutional neural networks can be trained and deployed on ESP32-class hardware to evaluate crop growth and detect objects directly from a photograph [8], [9]. Related work on tomato-leaf disease detection has shown that a TinyML pipeline running on an ESP32-S3 can classify foliar disease from an image in well under a second while remaining small enough to fit within the memory of a microcontroller [13].

From this review, it can be observed that most existing systems either automate irrigation purely from environmental sensor thresholds, or perform crop-image analysis as an isolated, offline exercise. Few systems combine both approaches into a single, low-cost, cloud-connected controller. This project therefore proposes a Vision-Assisted Smart Irrigation System that unifies soil, weather and intrusion sensing on the ESP32-S3 with a Firebase-mediated Edge Impulse image-classification pipeline, so that both real-time sensor readings and periodic crop-health assessment jointly inform the irrigation decision.

4. Problem Statement

4.1 Existing Problem

Irrigation scheduling in most small and medium farms is still performed manually or through fixed timers that do not respond to the actual moisture content of the soil, the presence of rainfall, or the visible health of the crop. This leads to overwatering, under watering, wasted electricity for pumping, and, in some cases, damage to crops from either water stress or fungal growth caused by excess moisture. In addition, fields are often left unattended for long periods, so animal intrusion or theft may go unnoticed until damage has already occurred.

4.2 Proposed Solution

This project proposes a Vision-Assisted Smart Irrigation System built around the ESP32-S3 microcontroller. A soil moisture sensor, a rain sensor and a DHT11 temperature-humidity sensor continuously evaluate the field conditions, while a PIR motion sensor watches for intrusion. The ESP32-S3 uses this sensor data, together with a periodically-updated crop-health flag obtained from an Edge Impulse image-classification model (delivered through a Firebase database), to decide when the 2-channel relay module should switch the water pump on or off. All readings, alerts and manual controls are exposed through the Blynk IoT dashboard, removing the need for a physical display.

4.3 Objectives

- To develop a low-cost, IoT-based smart irrigation system that reduces water wastage.
- To measure soil moisture, rainfall, temperature and humidity in real time.
- To detect intrusion or animal movement in the field using a PIR sensor.
- To automatically control a water pump using a relay based on sensor thresholds.
- To incorporate an image-based crop health check using Edge Impulse and Firebase.
- To provide fully remote monitoring and manual override through the Blynk IoT app.
- To design a compact, display-free hardware unit suitable for field deployment.

4.4 Expected Outcome

The expected outcome is a functioning, low-cost irrigation controller that waters a field only when the soil, weather and crop-health data indicate a genuine need for it, that alerts the user immediately to any intrusion, and that can be fully monitored and operated from a smartphone through the Blynk dashboard.

5. Methodology

5.1 System Approach

The proposed system follows a sensor-fusion IoT methodology in which field parameters are measured continuously by dedicated sensors, combined with a periodic crop-image classification result, and used by the ESP32-S3 to make an irrigation decision. The decision is executed through a relay-driven water pump and reported to the user through the Blynk IoT dashboard.

The overall methodology follows the sequence:

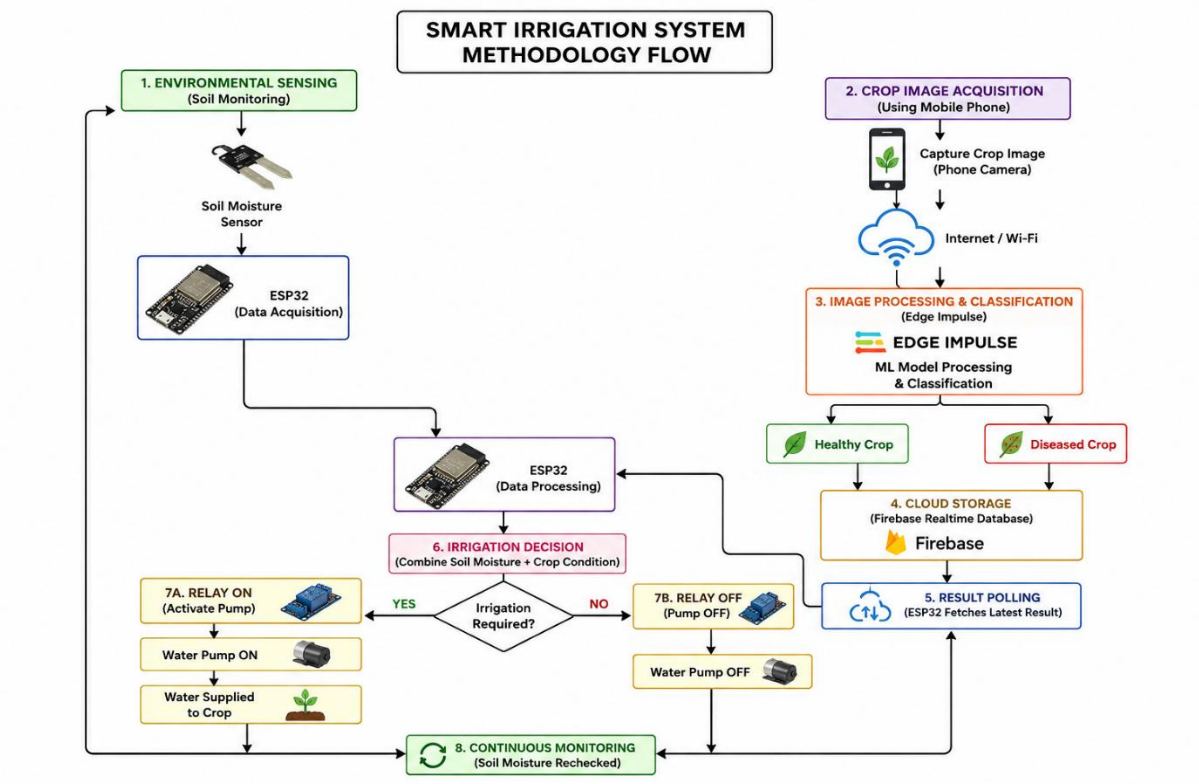

Fig (1): Methodology Flow of the System

5.2 Data Acquisition

The system uses the following sensors to acquire field information:

- Soil Moisture Sensor: Measures the volumetric water content of the soil (resistive probe).
- Rain Sensor: Detects the presence of rainfall or surface water.
- DHT11: Measures ambient temperature and relative humidity.
- PIR Motion Sensor: Detects the movement of a person or animal near the field.

The outputs of these sensors are connected to the analog/digital GPIO pins of the ESP32-S3.

5.3 Data Processing

The ESP32-S3 acts as the central processing unit. It periodically polls each connected sensor, converts the raw analog readings into meaningful engineering units (percentage soil moisture, °C, %RH) and evaluates whether the pre-defined watering thresholds have been met.

5.4 Actuation and Pump Control

Based on the combined evaluation of soil moisture, rain status and the crop-health flag retrieved from Firebase, the ESP32-S3 drives the 2-channel relay module, which in turn switches the 5V submersible water pump on or off. The relay's opto-isolated design protects the microcontroller from switching transients generated by the pump motor.

5.5 Wireless Data Transmission and Cloud Monitoring

The ESP32-S3 uses its built-in Wi-Fi module to connect to the internet and stream all sensor readings and the pump status to the Blynk IoT platform. This eliminates the need for a separate communication module and enables the user to view live gauges for soil moisture, rain, PIR, temperature and humidity, along with a pump-status indicator, from anywhere. Manual override buttons on the Blynk dashboard let the user switch the pump on/off directly and clear or act on motion-detection alerts.

5.6 Vision-Assisted Crop Health Monitoring

In parallel with the sensor-based automation, a crop image captured on a smartphone is uploaded to a server where an Edge Impulse-trained image-classification model evaluates the visible health of the crop (e.g., healthy, water-stressed, or diseased). The classification result is written to a Firebase Real-time Database. The ESP32-S3 periodically queries Firebase and factors the returned crop-health flag into its irrigation and alert logic, allowing the watering decision to be crop-specific rather than purely soil/weather-driven.

5.7 Real-Time Monitoring

The complete process; sensing, decision-making, actuation, cloud synchronization and crop-image inference; repeats continuously at a programmed interval, giving the user an up-to-date picture of both the field and the crop at all times.

6. System Design

6.1 System Architecture

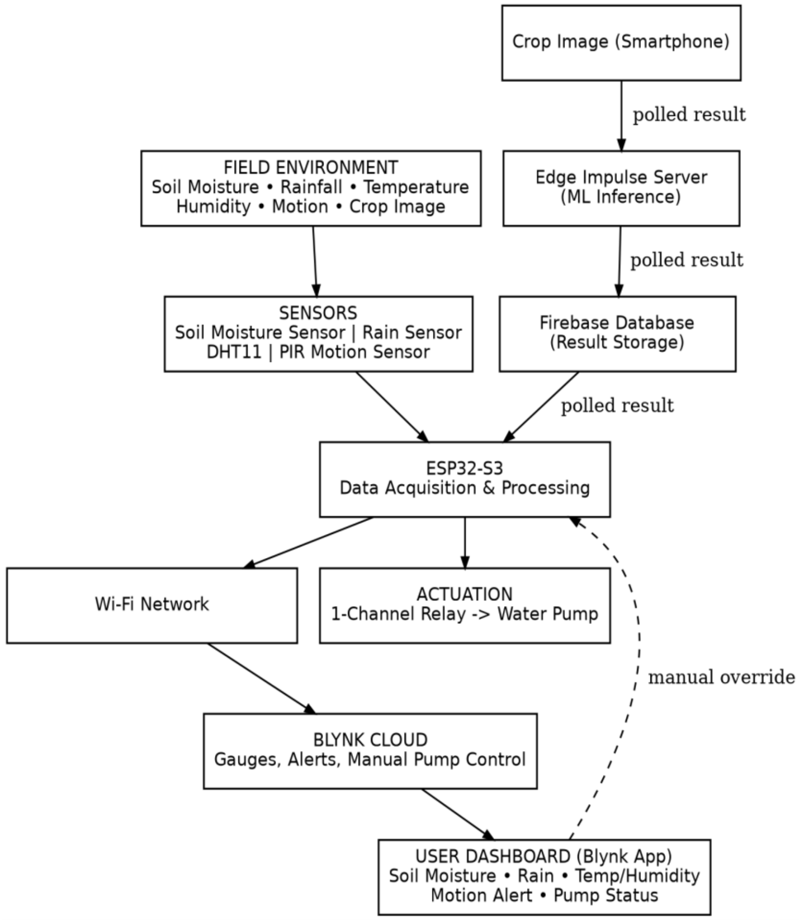

Fig (2) – System Architecture

6.2 Block Diagram

The block diagram below summarizes the wiring between each sensor/actuator and the ESP32-S3, and the two parallel data paths; the direct sensor-to-Blynk path, and the smartphone-image-to-Firebase-to-ESP32-S3 path.

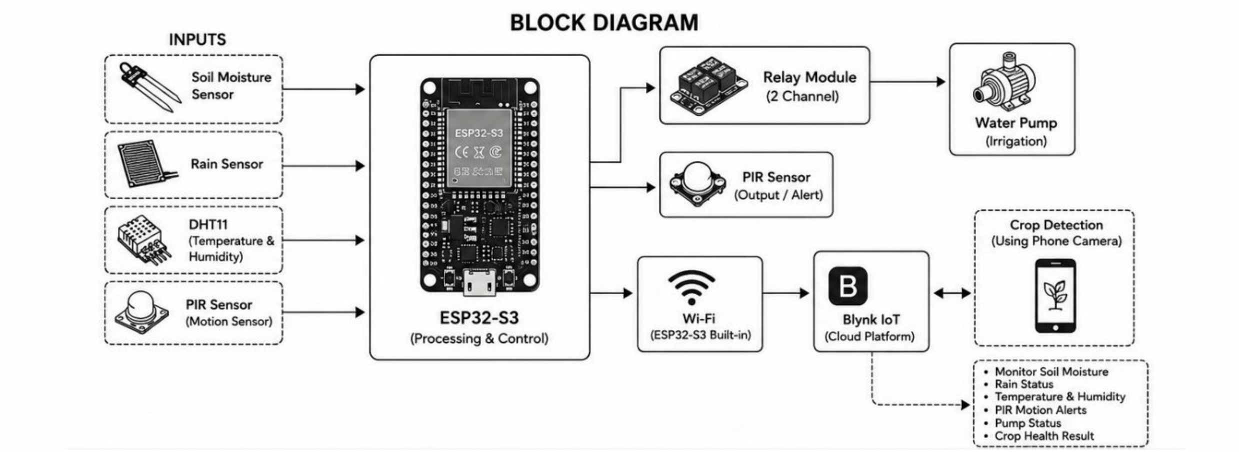

Fig (3) – Block Diagram of the System

6.3 Circuit Assembly

The hardware prototype was assembled on a breadboard, with the ESP32-S3, sensors, relay module, and water pump wired according to the pin configuration in Table (1).

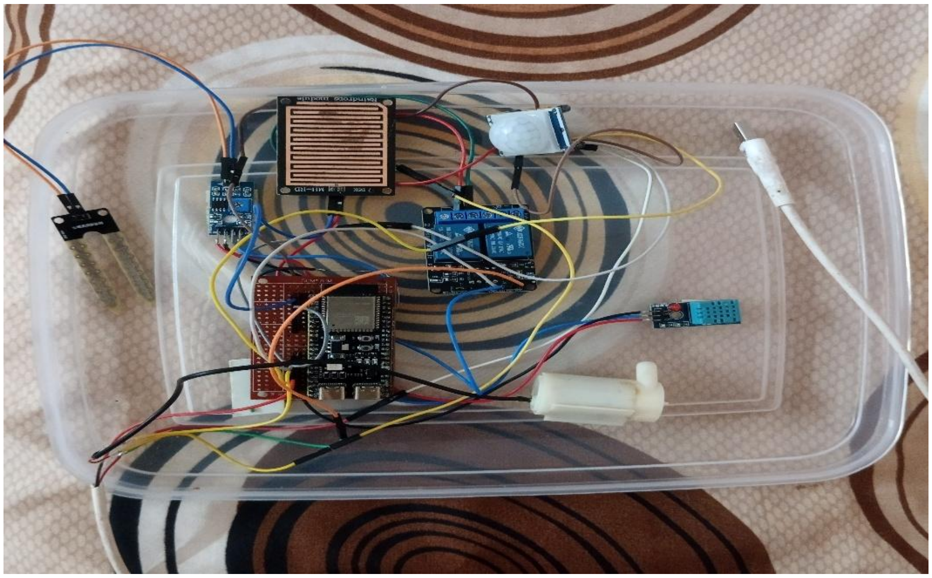

Fig (4) – Prototype Circuit

6.4 Pin Configuration

Component	Pin	ESP32-S3 Connection
Soil Moisture Sensor	AO	GPIO1
Soil Moisture Sensor	VCC / GND	3.3V / GND
Rain Sensor Module	AO	GPIO2
Rain Sensor Module	VCC / GND	3.3V / GND

DHT11	DATA	GPIO4
DHT11	VCC / GND	3.3V / GND
PIR Motion Sensor	OUT	GPIO5
PIR Motion Sensor	VCC / GND	5V / GND
Relay Module	IN	GPIO6
Relay Module	VCC / GND	5V / GND
Water Pump	+ / –	Relay NO – COM (5V supply)

Table (1) – Pin Configuration

7. Hardware Components

The proposed vision-assisted smart irrigation system consists of several hardware components that work together to sense field conditions, take irrigation decisions and actuate the water pump. The major components used in the system are described below.

7.1 ESP32-S3

The ESP32-S3-WROOM-1 (N16R8, Wi-Fi + BLE, USB-C) is the main microcontroller used in the proposed system. It collects data from the connected sensors, evaluates the irrigation logic, queries Firebase for the crop-health flag, and transmits all information to the Blynk IoT platform through its built-in Wi-Fi connectivity.

7.2 DHT11 Sensor

The DHT11 sensor measures ambient temperature (0–50 °C, ±2 °C) and relative humidity (20–90% RH, ±5%) and provides a digital output that can be connected directly to an ESP32-S3 GPIO pin.

7.3 Soil Moisture Sensor

A resistive soil moisture sensor is used to measure the volumetric water content of the soil. Its analog output is read through the ESP32-S3's ADC and converted into a moisture percentage.

7.4 Rain Sensor Module

The rain sensor module (LM393 comparator) provides both analog and digital outputs and is used to detect the presence of rainfall or surface water on its sensing plate, preventing unnecessary irrigation during or immediately after rain.

7.5 PIR Motion Sensor

The PIR (Passive Infrared) sensor detects the motion of a person or animal within its field of view and triggers an intrusion alert on the Blynk dashboard.

7.6 Relay Module

An opto-coupler isolated relay module is used to switch the water pump on and off under the control of the ESP32-S3, protecting the microcontroller from the switching transients of the pump motor.

7.7 Water Pump

A 3-6V/5V submersible mini water pump (80–120 L/hr. flow rate) is used to deliver water to the field when the relay is energized.

7.8 5V Power Adaptor

A 5V DC power adaptor supplies the main power rail for the water pump.

7.9 Zero PCB and Wires

A zero PCB and a set of 20 cm wires are used to assemble and test the circuit without permanent soldering, allowing the layout to be modified easily during development.

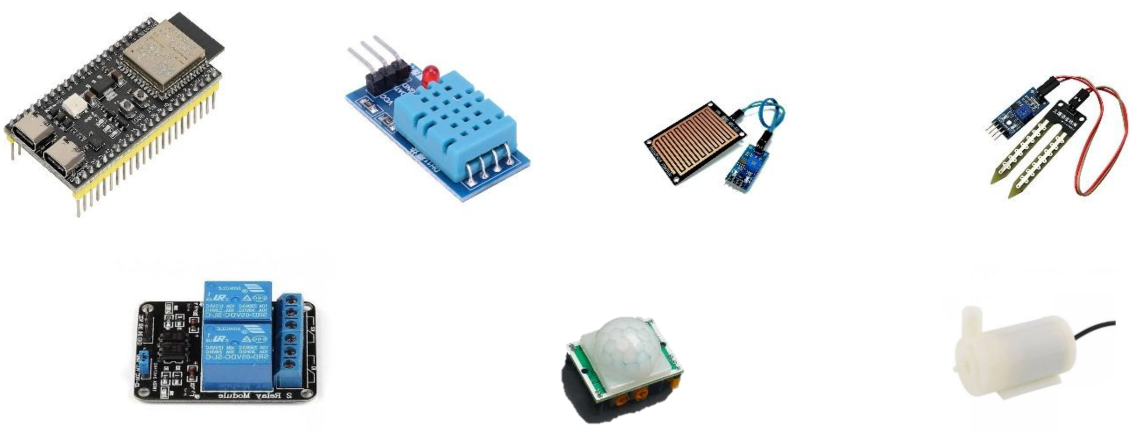

Fig (5) – Hardware Components used in the System

7.10 Bill of Materials

The table below lists the components actually used for the prototype, along with their approximate procurement cost.

Component	Specification	Qty	Cost (₹)
ESP32-S3-WROOM-1 (N16R8)	N16R8, Wi-Fi + BLE, USB-C	1	550 – 750
DHT11	Temp: 0–50 °C (±2 °C), Humidity: 20–90% RH (±5%), 3-pin module	1	60 – 90
Relay Module	5V, 1-channel, 10A/250VAC, opto-coupler isolated	1	60 – 80

PIR Motion Sensor	HC-SR501 type, digital output (used for intrusion detection)	1	50 – 70
Rain Sensor Module	LM393 comparator, analog + digital out	1	60 – 90
Water Pump	3-6V/5V submersible mini pump, 80–120 L/hr. flow rate	1	100 – 180
5V Adaptor	5V DC power adapter	1	150 – 200
Soil Moisture Sensor	Resistive soil moisture sensor module.	1	80 – 120
Wires	20 cm, set of 40	1 set	60 – 80
Zero PCB	Standard 2.54 mm (0.1 inch)	1 set	25-40

Table (2) – Bill of Materials

Approximate Total Cost: ₹ 1,210 – 1,720

8. Software Requirements

- Arduino IDE – used to write, compile and upload the program to the ESP32-S3, and to test/debug it through the Serial Monitor.
- ESP32 Board Package – installed in Arduino IDE to provide board definitions and compiler support for the ESP32-S3.

8.1 Required Libraries

Library	Purpose
DHT Sensor Library	Reading temperature and humidity from DHT11
Adafruit Unified Sensor	Support library for sensor interfacing
WiFi Library	Connecting the ESP32-S3 to the internet
Blynk Library	Communication between ESP32-S3 and the Blynk IoT app
Firebase ESP32 Client Library	Reading/writing the crop-health flag from Firebase
Edge Impulse Arduino Library / SDK	Running the crop image-classification model on the server/edge device

Table (3) – Required Software Libraries

- Blynk IoT Platform – cloud-based dashboard used to display live gauges for soil moisture, rain, PIR, temperature and humidity, the pump-status indicator, and manual override buttons.

- Firebase Real-time Database – stores the crop-health classification result generated by Edge Impulse for the ESP32-S3 to consult.
- Edge Impulse Studio – used to collect crop images, train and export the image-classification model used for crop health inference.
- Serial Monitor – used in Arduino IDE for testing and debugging sensor readings and system messages in real time.

9. Working Principle

The working principle of the proposed system is based on the continuous acquisition of field data, an on-board decision process on the ESP32-S3, and remote visualization and control through the Blynk IoT app.

The system begins by initializing the ESP32-S3 and the connected sensors, and by establishing a Wi-Fi connection. The soil moisture sensor, rain sensor and DHT11 continuously report the condition of the soil and the surrounding air, while the PIR sensor watches for any motion in the field.

The ESP32-S3 evaluates the soil-moisture reading against a pre-set threshold; if the soil is dry and the rain sensor does not indicate rainfall, the microcontroller checks the latest crop-health flag stored in Firebase (updated whenever a new smartphone image has been classified by the Edge Impulse model). If the combined sensor and crop-health evaluation indicates that watering is required, the ESP32-S3 energizes the relay, which switches the water pump on; otherwise, the pump remains off, preventing unnecessary water use.

If the PIR sensor detects motion, the ESP32-S3 immediately raises an alert on the Blynk dashboard so that the user can respond to a possible intrusion or animal movement in the field. At any point, the user may also use the Blynk app's manual buttons to switch the pump on or off directly, overriding the automatic logic, based on the live readings shown on the dashboard.

Separately, whenever a smartphone photograph of the crop is taken, it is sent to a server for inference using the Edge Impulse model; the resulting classification (e.g., healthy / stressed / diseased) is written to Firebase, which the ESP32-S3 polls at a regular interval so that the vision-based assessment stays incorporated into the irrigation decision.

This process; sensing, decision-making, actuation, cloud synchronization and periodic image-based crop assessment, repeats continuously, giving the user an up-to-date, remotely accessible picture of both the field and the crop at all times.

10. Simulation and Testing

10.1 Simulation Environment

The proposed system was simulated using Wokwi, an online electronics simulation platform, to verify the ESP32-S3 program, sensor connections and the basic irrigation logic before implementing the system on physical hardware.

10.2 Simulation Setup

The Wokwi simulation included the ESP32-S3-WROOM-1, a relay module, a DHT22 sensor (used as a substitute for the DHT11, since the exact part was unavailable in the simulator's component library), a PIR motion sensor, and two potentiometers used to emulate the variable analog outputs of the soil moisture sensor and the rain sensor, since neither module was available in the Wokwi environment. An LED with a current-limiting resistor was used in place of the water pump to visually indicate the relay/pump state.

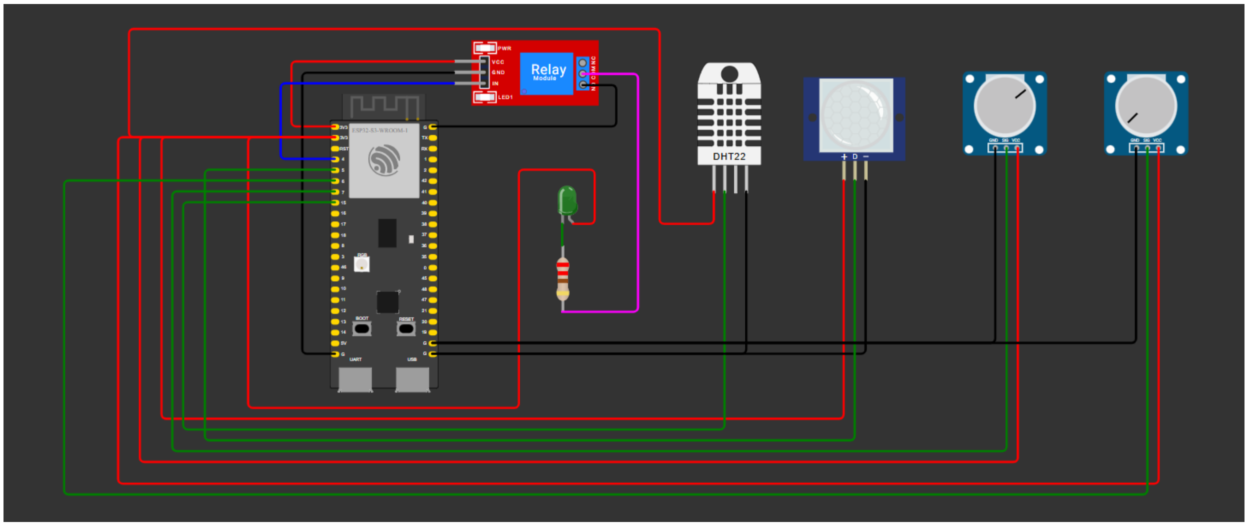

Fig (6) – Simulation Circuit in Wokwi

10.3 Simulation Procedure

The ESP32-S3 program was uploaded and executed in Wokwi. The potentiometers were varied to simulate different soil-moisture and rainfall conditions, and the PIR sensor was toggled to simulate motion detection. The resulting sensor readings, relay state and decision logic were observed through the Serial Monitor to verify correct operation.

10.4 Simulation Result

The Wokwi simulation successfully verified sensor interfacing, ADC-based moisture/rain readings, PIR-based motion detection, and correct switching of the relay/pump according to the programmed thresholds.

```
ESP-ROM:esp32s3-20210327
Build:Mar 27 2021
rst:0x1 (POWERON),boot:0x8 (SPI_FAST_FLASH_BOOT)
SPIWP:0xee
mode:DIO, clock div:1
load:0x3fce3808,len:0x370
load:0x403c9700,len:0x900
load:0x403cc700,len:0x2364
entry 0x403c98ac

=====================================
       SMART IRRIGATION SYSTEM
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-------------------------------------
Soil Moisture : 1000
Rain Sensor   : 1000
PIR Motion    : NOT DETECTED
Temperature   : 28.00 °C
Humidity      : 65.00 %
Soil is dry → PUMP ON
-------------------------------------
Soil Moisture : 1100
Rain Sensor   : 900
PIR Motion    : DETECTED
Temperature   : 28.00 °C
Humidity      : 65.00 %
Soil is dry → PUMP ON
-------------------------------------
Soil Moisture : 2600
Rain Sensor   : 900
PIR Motion    : NOT DETECTED
Temperature   : 28.00 °C
Humidity      : 65.00 %
Soil is wet → PUMP OFF
-------------------------------------
Soil Moisture : 1200
Rain Sensor   : 3000
PIR Motion    : NOT DETECTED
Temperature   : 29.00 °C
Humidity      : 70.00 %
Rain detected → PUMP OFF
```

Fig (7) – Simulation Result

11. Implementation

11.1 Hardware Implementation

The system was implemented using an ESP32-S3, DHT11, soil moisture sensor, rain sensor, PIR sensor, relay module, buck converter, 5V adaptor and a submersible water pump. The components were connected according to the pin configuration in Table (1) and assembled using a breadboard and wires.

11.2 Sensor Integration

The sensors were connected to the ESP32-S3 to collect soil moisture, rainfall, temperature, humidity and motion data. The ESP32-S3 reads and processes the sensor outputs at regular intervals and combines them with the crop-health flag retrieved from Firebase.

11.3 Software Implementation

The ESP32-S3 was programmed using Arduino IDE. The required libraries were included for sensor interfacing, Wi-Fi communication, Blynk integration, and Firebase communication. The program was uploaded to the ESP32-S3 and tested using the Serial Monitor.

11.4 Blynk Implementation

The processed sensor data and pump status are transmitted through Wi-Fi to the Blynk IoT platform. The Blynk dashboard is configured with live gauges for soil moisture, rain, PIR, temperature and humidity, a pump-status indicator, and manual override buttons for the pump and intrusion alerts.

11.5 Edge Impulse and Firebase Implementation

A crop image dataset was used to train an image-classification model in Edge Impulse Studio. The trained model is deployed on a server that accepts smartphone-captured crop images, runs inference, and writes the resulting crop-health label to a Firebase Real-time Database, which the ESP32-S3 queries periodically over Wi-Fi.

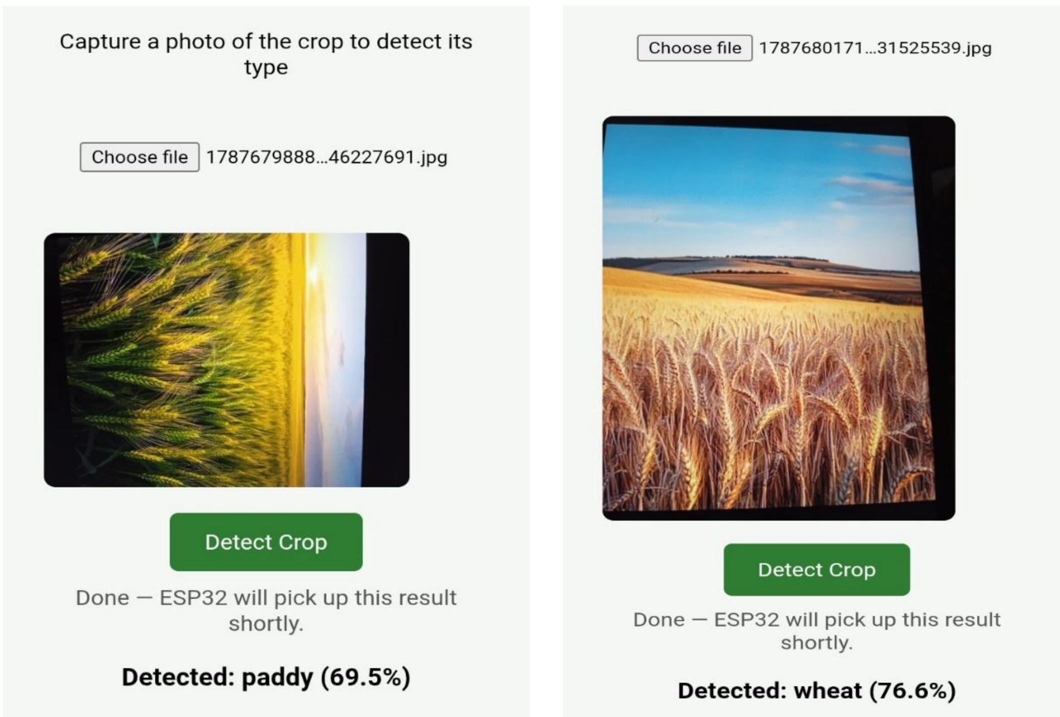

Fig (8) – Crop Image Detection

12. Results and Discussion

12.1 Experimental Results

The implemented system was tested using the connected sensors to obtain real-time field readings. The ESP32-S3 successfully collected and processed the soil moisture, rain, temperature, humidity and PIR data, and correctly triggered the relay/pump according to the programmed thresholds.

12.2 Sensor Readings

The system provided readings for soil moisture, rainfall status, temperature, humidity and motion detection. The readings were observed through the Serial Monitor and the Blynk dashboard, and the pump switched on only when the soil-moisture reading fell below the configured threshold and no rainfall was detected.

12.3 Blynk Dashboard Results

The processed sensor data was transmitted to the Blynk IoT platform through Wi-Fi and displayed on the dashboard, showing live gauges for soil moisture, rain, PIR, temperature and humidity along with the pump-status indicator, enabling fully remote monitoring of the field.

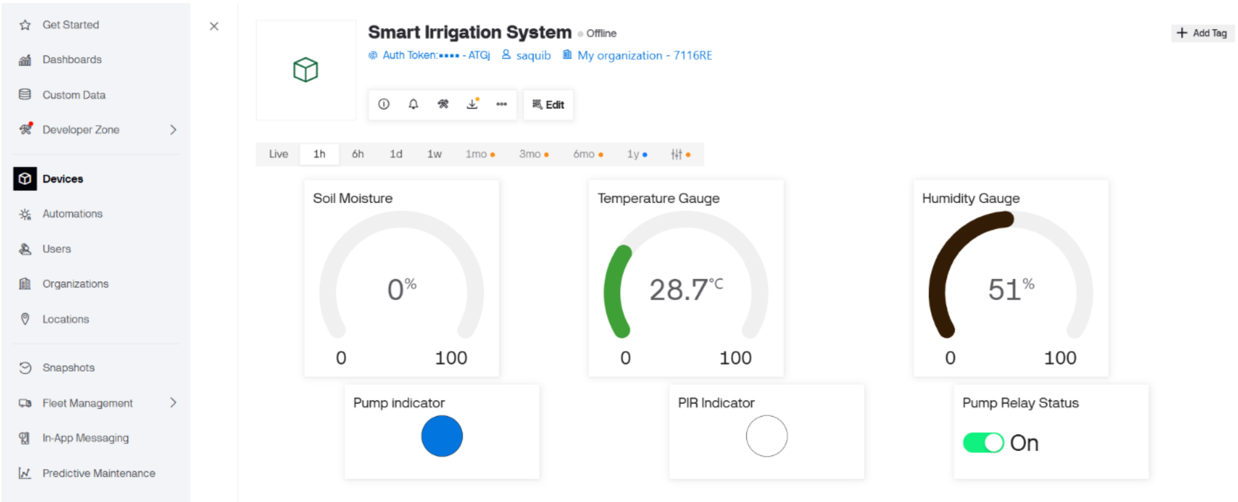

Fig (9) – Blynk IoT Dashboard for Real-Time Field Monitoring

12.4 Discussion

The results demonstrate that the ESP32-S3-based system can successfully integrate multiple environmental sensors, a relay-driven pump, and a Firebase-mediated Edge Impulse crop-health check into a single, remotely monitored irrigation controller. The combination of sensor-driven automation, cloud dashboard visualization and vision-based crop assessment provides a convenient and water-efficient method for crop-specific irrigation.

During testing, the soil-moisture-and-rain threshold logic responded correctly to changing field conditions: the pump switched ON only when the soil moisture reading fell below the configured dry threshold and no rainfall was simultaneously detected, and switched OFF as soon as either condition changed. This behavior matches the automation pattern reported in the reviewed literature [1], [5], [6], confirming that a low-cost sensor-and-relay approach is sufficient to prevent the overwatering and under-watering associated with manual or timer-based irrigation. The DHT11 and PIR readings were also found to update reliably on the Blynk dashboard at the configured polling interval, with no noticeable lag between a sensor event (such as motion in the field) and the corresponding alert appearing on the app.

The removal of a physical display in favor of the Blynk IoT app was found to work well in practice: since all gauges, the pump-status indicator, and the ON/OFF buttons are available remotely, the user was able to monitor and, when needed, override the automatic pump behavior without being physically present near the enclosure. This is a meaningful improvement over several of the systems discussed in the literature survey, most of which limit remote access to passive monitoring rather than also allowing the user to act directly on the pump or on a motion alert from the dashboard itself [2], [3], [4].

13. Innovation

The proposed system introduces a novel direction in IoT-based irrigation by combining multi-parameter environmental sensing, edge-level decision-making, intrusion detection, real-time cloud connectivity and image-based crop-health assessment within a single, low-cost and display-free platform. Rather than functioning only as a threshold-based irrigation timer, the system is designed as an integrated field-intelligence node in which soil moisture, rainfall, temperature, humidity and motion are acquired simultaneously and processed at the ESP32-S3, while a parallel Edge Impulse image-classification pipeline evaluates the visible health of the crop and feeds that result back into the irrigation decision through Firebase.

A key innovative aspect is this fusion of conventional sensor thresholds with a lightweight, smartphone-driven computer-vision check, so that watering is not only weather- and soil-aware but also crop-specific. The architecture can further be extended toward on-device (TinyML) inference, multi-zone irrigation, adaptive thresholds and yield-prediction analytics without changing the fundamental hardware structure.

14. Advantages

- Low-cost and compact design with no dedicated physical display.
- Fully remote monitoring and control through the Blynk IoT dashboard.
- Reduces water wastage by watering only when soil, weather and crop-health conditions genuinely require it.
- Provides real-time intrusion/motion alerts from the field.
- Crop-specific irrigation decisions through image-based crop-health assessment.
- Wi-Fi communication without requiring a separate network module.
- Easy to expand with additional sensors or irrigation zones.

15. Limitations

- DHT11 provides relatively basic temperature and humidity accuracy.
- The capacitive soil moisture sensor requires calibration for different soil types.
- The rain sensor mainly indicates the presence of water and requires suitable outdoor placement.
- The Edge Impulse crop-health classification depends on the quality and lighting of the smartphone image and requires an internet connection to reach the server and Firebase.
- Wi-Fi-based cloud monitoring depends on network availability at the field location.
- Sensor placement and environmental exposure can affect measurement accuracy.

16. Future Scope

- Run the Edge Impulse crop-health model directly on-device (TinyML) using an ESP32-CAM to remove dependence on a remote server.
- Add multi-zone irrigation control with independent relays and pumps.
- Use more accurate soil-moisture, temperature and humidity sensors.
- Add GPS for field/zone location tagging.
- Store historical sensor and irrigation data for trend analysis and yield correlation.
- Implement solar power to make the unit fully self-sufficient in the field.

17. Mathematical Modelling and Equations

The proposed system involves the acquisition, processing and transmission of field data using multiple sensors connected to the ESP32-S3. For a 12-bit ADC, the digital reading of an analog sensor (soil moisture or rain sensor) corresponds to an input voltage given by:

$$v\_out = (ADC / 4095) \times V\_ref$$

where ADC is the digital value obtained from the ESP32-S3 and V_ref is the reference voltage.

The soil moisture percentage is estimated from the raw ADC reading using the calibration values obtained for a fully dry (ADC_dry) and a fully saturated (ADC_wet) soil sample:

$$\text{Soil Moisture (\%)} = [(ADC\_dry - ADC) / (ADC\_dry - ADC\_wet)] \times 100$$

The rain sensor output is interpreted using a predefined threshold to determine whether rain or water is detected, and can be represented as:

$$R = 1, \text{Rain Detected} \qquad R = 0, \text{No Rain Detected}$$

where R represents the rain status.

The overall pump-actuation decision, P, combines the soil-moisture condition, the rain status and the crop-health flag (C, obtained from the Edge Impulse/Firebase pipeline, where C = 1 indicates a healthy crop requiring normal irrigation and C = 0 indicates a condition where irrigation should be withheld) as follows:

$$P = 1, \text{if (Soil Moisture < Threshold) AND } (R = 0) \text{ AND } (C = 1)\text{; otherwise } P = 0$$

The volume of water delivered during a single irrigation cycle can be estimated from the pump's rated flow rate, Q, and the time for which the relay remains energized, t:

$$V = Q \times t$$

These mathematical relationships provide the basis for converting raw sensor outputs into the meaningful field information used by the irrigation decision logic.

18. Applications

- Smart Farms: Automated, crop-specific irrigation for open fields and orchards.
- Greenhouses and Polyhouses: Precise moisture control in a controlled environment.
- Home and Terrace Gardening: Low-cost automated watering with remote monitoring.
- Educational Institutions: Demonstration platform for IoT, embedded systems and edge AI.
- Research and Precision Agriculture: A foundation for studying sensor fusion and vision-assisted irrigation strategies.

19. Conclusion

The Vision-Assisted Smart Irrigation System demonstrates how an ESP32-S3, a set of low-cost environmental sensors, and a lightweight image-classification pipeline built on Edge Impulse and Firebase can be integrated into a single, water-efficient irrigation platform. By combining real-time soil, weather and intrusion sensing with a periodic, crop-specific image-based health check, and by exposing all monitoring and control through the Blynk IoT app, the system provides a practical, low-cost and scalable solution for smart, crop-specific water management. Its modular design also allows future improvements in on-device inference, multi-zone control and long-term data analytics.

A key feature of the proposed system is the integration of crop-specific image analysis. By using a lightweight image-classification model developed through Edge Impulse and supported by Firebase, the system can periodically analyze images of the crop and provide additional information about its condition. This adds another layer of intelligence to the irrigation process, as decisions can consider not only the moisture level of the soil but also the visual condition of the crop. Such an approach can help make irrigation more responsive to the actual requirements of the plants.

Thus, the Vision-Assisted Smart Irrigation System represents a step toward more intelligent, automated, and sustainable agriculture. With further development and field-level testing, this approach has the potential to become a scalable solution for precision agriculture, particularly in situations where conserving water, reducing manual effort, and improving crop monitoring are important.

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